MATHEMATICS — B.TECH CSE

SEMESTER I • SYLLABUS INDEX

UNIT 1 • ONE VARIABLE CALCULUS

- Introduction to One Variable Calculus, Limits & Standard Limits
- Continuity and Differentiability of Single Variable Functions
- Successive Differentiation of Standard Functions
- Rolle's Theorem and Applications
- Lagrange's Mean Value Theorem and Applications
- Taylor's Theorem for One Variable Functions & Numerical Applications

UNIT 2 • TWO VARIABLE CALCULUS

- Functions of Two Variables, Limits and Continuity
- Partial Derivatives and Higher Order Partial Derivatives
- Total Derivative and Chain Rule
- Euler's Theorem for Homogeneous Functions
- Taylor Series for Functions of Two Variables
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UNIT 3 • APPLICATION OF DIFFERENTIAL CALCULUS

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UNIT 4 • INTEGRAL CALCULUS

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